

Seminario de Investigación Económica

Structural Vector Autoregressions¹

Weeks 3 and 4

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¹Slides based on Ambrogio Cesa-Bianchi (Bank of England) and Alejandro Viccondoa (U. Católica de Chile).

Structural VAR

- Remember the VAR(1) expressed in matrix form

$$x_t = \Phi x_{t-1} + u_t$$

where $u_t \sim \mathcal{N}(0, \sigma_u^2)$. This is the **reduced-form VAR**

- The **structural VAR** can be defined as

$$x_t = \Phi x_{t-1} + B\varepsilon_t$$

where B is a (2×2) matrix and $\varepsilon_t \sim \mathcal{N}(0, I)$. In other words we assume

$$\varepsilon_t = (\varepsilon'_{1t}, \varepsilon'_{2t})' \sim \mathcal{N}(0, I_2)$$

where

$$\text{Var}(\varepsilon_t) = \Sigma_\varepsilon = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \text{Corr}(\varepsilon_t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Why is it called 'structural' VAR?

- Go back to our bivariate structural VAR(1)

$$\begin{bmatrix} x_{1t} \\ x_{2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} x_{1t-1} \\ x_{2t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

- The structural VAR can be thought of as a description of the true structure of the economy
 - E.g.: an approximation of the structure of a DSGE model

The model is structural in that the elements of ε_t are mutually uncorrelated and have clear interpretations in terms of an underlying economic model.

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- The structural shocks are shocks with a well-defined economic interpretation
 - E.g.: TFP shocks or monetary policy shocks
 - As $\varepsilon_t \sim \mathcal{N}(0, I_2)$ we can move one shock keeping the other shocks fixed
 - That is: we can focus on the causal effect of one shock at the time

Structural VARs can answer many interesting questions...

- Go back to our bivariate structural VAR(1). To make a concrete example, assume that
 - x_{1t} and x_{2t} are output growth (y_t) and the policy rate (r_t), both demeaned
 - ε_{1t} and ε_{2t} are a demand shock (ε_t^{Demand}) and a monetary policy shock (ε_t^{MonPol})
 - B is known (we'll get back to this in a few slides)

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 - (We can add additional variables and look at other shocks: aggregate supply, oil price,...)

... but the estimation of structural VARs is tricky

- **Problem** The structural shocks ε_t are unobserved. How can we estimate B ?

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- Best we can do is to 'bundle' the ε_t into a single object:

$$u_t = B\varepsilon_t \Rightarrow \begin{bmatrix} u_{1t} \\ u_{2t} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix} \Rightarrow \begin{cases} u_{yt} = b_{11}\varepsilon_t^{Demand} + b_{12}\varepsilon_t^{MonPol} \\ u_{rt} = b_{21}\varepsilon_t^{Demand} + b_{22}\varepsilon_t^{MonPol} \end{cases}$$

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- Why is this useful? The VAR becomes

$$x_t = \Phi x_{t-1} + u_t$$

- Now we can estimate Φ and u_t with OLS (where u_t will be OLS residuals)

The reduced-form VAR

- This alternative formulation of the VAR is called the **reduced-form VAR** representation (we reviewed this last slides)

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- In matrix form

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- Or as a system of linear equations

$$\begin{cases} y_t = \phi_{11}y_{t-1} + \phi_{12}r_{t-1} + u_{yt} \\ r_t = \phi_{21}y_{t-1} + \phi_{22}r_{t-1} + u_{rt} \end{cases}$$

The reduced-form covariance matrix

- A key object of interest in VARs is the covariance matrix of the **reduced-form residuals**

$$\Sigma_u = \begin{bmatrix} \sigma_y^2 & \sigma_{yr}^2 \\ \sigma_{yr}^2 & \sigma_r^2 \end{bmatrix}$$

- Differently from the structural shocks (which are orthogonal), **the reduced-form residuals are correlated among each other**

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- Differently from the structural shocks (which are orthogonal), **the reduced-form residuals are correlated among each other**
- This is because the elements of u_t inherit all the contemporaneous relations among the endogenous variables x_t
 - To see that, remember how the reduced form residuals are defined

$$\begin{cases} u_{yt} = b_{11}\varepsilon_t^{Demand} + b_{12}\varepsilon_t^{MonPol} \\ u_{rt} = b_{21}\varepsilon_t^{Demand} + b_{22}\varepsilon_t^{MonPol} \end{cases}$$

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- To make causal statements (e.g. the effects on y_t of a shock to ε_t^{MonPol}) we need to find a way to recover B
- This is the essence of **identification** in VARs

The Identification Problem

Back to our reduced form VAR

- We have seen above that with OLS we can only estimate the reduced-form VAR (and not the structural VAR)
- Assume we already have an OLS estimate of $\hat{\Phi}$ and $\hat{u}_t$:

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} u_{yt} \\ u_{rt} \end{bmatrix}$$

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- **Question** What is the effect of a monetary policy shock on GDP growth?
- Unfortunately, the reduced form innovations (u_{yt} or u_{rt}) are not going to help us in answering the question

Reduced-form VARs do not tell us anything about causality

- To see that, assume that the 'true' (and unobserved) model of the economy is given by

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- An increase in u_{rt} is not a monetary policy shock

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- How to know whether is [1] or [2]? This is the very nature of the **identification problem**

The identification problem

- The identification problem consists in finding a mapping from the reduced form VAR to its structural counterpart

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- To do that, we can exploit the relation between reduced form and structural innovations to write

$$\Sigma_u = \mathbb{E} [u_t u_t'] = \mathbb{E} [B\varepsilon_t (B\varepsilon_t)'] = B\mathbb{E}(\varepsilon_t \varepsilon_t')B' = B\Sigma_\varepsilon B' = BB'$$

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- The identification problem simply boils down to finding a B matrix that satisfies $\Sigma_u = BB'$
- Unfortunately this is not as easy as it sounds. Why?
 - Hint: There are infinite combinations of B that give you the same Σ_u

The identification problem (cont'd)

- Think of $\Sigma_u = BB'$ as a system of equations

$$\begin{bmatrix} \sigma_y^2 & \sigma_{yr}^2 \\ - & \sigma_r^2 \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix}$$

- Can be rewritten as

$$\begin{cases} \sigma_y^2 = b_{11}^2 + b_{12}^2 \\ \sigma_{yr}^2 = b_{11}b_{21} + b_{12}b_{22} \\ \sigma_{yr}^2 = b_{11}b_{21} + b_{12}b_{22} \\ \sigma_r^2 = b_{21}^2 + b_{22}^2 \end{cases}$$

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- **Problem** Because of the symmetry of the Σ_u matrix, the second and the third equation are identical
- We are left with 4 unknowns (the elements of B) but only 3 equations!

Structural Dynamic Analysis

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- What can we do with that?
 - Quantify the dynamic effect of a shock over time $\Rightarrow$ **Impulse responses**
 - Quantify how important a shock is in explaining the variation in the endogenous variables (on average) $\Rightarrow$ **Forecast error variance decomposition**
 - Quantify how important a shock was in driving the behavior of the endogenous variables in a specific time period in the past $\Rightarrow$ **Historical decompositions**

- We'll turn to this structural dynamic analysis next

Impulse response functions

- Impulse response functions (IR) answer the following question:

What is the response over time of the VAR's endogenous variables to an innovation in the structural shocks, assuming that the other structural shocks are kept to zero?

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- IR allow to single out the effect of a shock (e.g. its impact and persistence) keeping all else equal
- **Example** What is the impact of a monetary policy shock to GDP?

How to compute impulse response functions

- Consider our simple bivariate VAR

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

- Define a 2×1 impulse selection vector (s) that takes value of one for the structural shock that we want to consider.
- For example, to compute the *IR* to the demand shock, define s as:

$$s = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- The impulse responses to ε_t^{Demand} can be easily computed with the following equation

$$x_t = \Phi x_{t-1} + Bs$$

How to compute impulse response functions (cont'd)

- The IR can be computed recursively as follows

$$\begin{cases} IR_t = Bs & \text{for } t = 0 \\ IR_t = \Phi IR_{t-1} & \text{for } t = 1, \dots, h \end{cases}$$

- Note that the impact response is simply given by the elements of the impact matrix B selected by s ...

$$\begin{bmatrix} IR_0^y \\ IR_0^r \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix}$$

- ... while the responses at longer horizons are given the transition matrix

$$\begin{bmatrix} IR_t^y \\ IR_t^r \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} IR_{t-1}^y \\ IR_{t-1}^r \end{bmatrix}$$

The companion matrix [Back to basics]

- So far, we considered simple VAR(1) specifications. But what to do if the VAR has $p > 1$?
- Every VAR(p) can be written as a VAR(1) using the **companion representation**
 - For example, take a VAR(2)

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11}^1 & \phi_{12}^1 \\ \phi_{21}^1 & \phi_{22}^1 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} \phi_{11}^2 & \phi_{12}^2 \\ \phi_{21}^2 & \phi_{22}^2 \end{bmatrix} \begin{bmatrix} y_{t-2} \\ r_{t-2} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

- Re-write the VAR(2) as

$$\begin{bmatrix} y_t \\ r_t \\ y_{t-1} \\ r_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_{11}^1 & \phi_{12}^1 & \phi_{11}^2 & \phi_{12}^2 \\ \phi_{21}^1 & \phi_{22}^1 & \phi_{21}^2 & \phi_{22}^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \\ y_{t-2} \\ r_{t-2} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & 0 & 0 \\ b_{21} & b_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \\ 0 \\ 0 \end{bmatrix}$$

The companion matrix [Back to basics]

- So far, we considered simple VAR(1) specifications. But what to do if the VAR has $p > 1$?
- Every VAR(p) can be written as a VAR(1) using the **companion representation**
 - For example, take a VAR(2)

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11}^1 & \phi_{12}^1 \\ \phi_{21}^1 & \phi_{22}^1 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} \phi_{11}^2 & \phi_{12}^2 \\ \phi_{21}^2 & \phi_{22}^2 \end{bmatrix} \begin{bmatrix} y_{t-2} \\ r_{t-2} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

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- Which is a VAR(1) where $\tilde{\Phi}$ is the **companion matrix**

$$\tilde{x}_t = \tilde{\Phi} \tilde{x}_{t-1} + \tilde{B} \varepsilon_t$$

Forecast error variance decompositions

- Forecast error variance decompositions (VD) answer the following question:

What portion of the variance of the VAR's forecast errors (at a given horizon h) is due to each structural shock?

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- VD provide information about the relative importance of each structural shock in affecting the forecast error variance of the VAR's endogenous variables
- **Example** What is the (average) importance of demand shocks in driving GDP forecast errors?

How to compute forecast error variance decompositions

- The forecast error of a variable at horizon $t + h$ is the change in the variable that couldn't have been forecast between $t - 1$ and $t + h$ due to the realization of the structural shocks.

- For example, at $h = 0$ we can compute the forecast error as

$$x_t - \mathbb{E}[x_t | \Omega_{t-1}] = \Phi x_{t-1} + B\varepsilon_t - \Phi x_{t-1} = B\varepsilon_t$$

- Where as usual, Ω_t is the information set at time t . At $h = 1$, we have

$$\begin{aligned} x_{t+1} - \mathbb{E}[x_{t+1} | \Omega_{t-1}] &= \Phi x_t + B\varepsilon_{t+1} - \Phi^2 x_{t-1} = \\ &= \Phi(\Phi x_{t-1} + B\varepsilon_t) + B\varepsilon_{t+1} - \Phi^2 x_{t-1} = \Phi B\varepsilon_t + B\varepsilon_{t+1} \end{aligned}$$

- So, in general we have

$$FE_{t+h} = x_{t+h} - \mathbb{E}[x_{t+h} | \Omega_{t-1}] = \sum_{i=0}^h \Phi^{h-i} B\varepsilon_{t+i}$$

- What is the variance of FE_{t+h} ?

Basic properties of the variance [Back to basics]

- If X is a random variable x and a is a constant

- $Var(x + a) = Var(x)$

- $Var(ax) = a^2 Var(x)$

- If Y is a random variable and b is a constant

- $Var(aX + bY) = a^2 Var(x) + b^2 Var(Y) + 2abCov(X, Y)$

- Since the structural errors are independent, it follows that $Cov(\varepsilon_{t+1}^{Demand}, \varepsilon_{t+1}^{MonPol}) = 0$

How to compute forecast error variance decompositions (cont'd)

- For simplicity consider $h = 0$, namely

$$\text{Var}(FE_t) = \text{Var}(x_t - \mathbb{E}[x_t | \Omega_{t-1}]) = \text{Var}(B\varepsilon_t)$$

- Recalling that $\text{Var}(\varepsilon_t) = I_2$ and that the structural shocks are orthogonal to each other, the variance of the forecast error can be computed as

$$\text{Var}(y_t - \mathbb{E}[y_t | \Omega_{t-1}]) = b_{11}^2 \text{Var}(\varepsilon_t^{\text{Demand}}) + b_{12}^2 \text{Var}(\varepsilon_t^{\text{MonPol}}) = b_{11}^2 + b_{12}^2$$

$$\text{Var}(r_t - \mathbb{E}[r_t | \Omega_{t-1}]) = b_{21}^2 \text{Var}(\varepsilon_t^{\text{Demand}}) + b_{22}^2 \text{Var}(\varepsilon_t^{\text{MonPol}}) = b_{21}^2 + b_{22}^2$$

- What portion of the variance of the forecast error at $h = 0$ is due to each structural shock?

$$\underbrace{\begin{cases} VD_{y_0}^{\varepsilon^{\text{Demand}}} = \frac{b_{11}^2}{b_{11}^2 + b_{12}^2} \\ VD_{y_0}^{\varepsilon^{\text{MonPol}}} = \frac{b_{12}^2}{b_{11}^2 + b_{12}^2} \end{cases}}_{\text{This sums up to 1}} \quad \underbrace{\begin{cases} VD_{r_0}^{\varepsilon^{\text{Demand}}} = \frac{b_{21}^2}{b_{21}^2 + b_{22}^2} \\ VD_{r_0}^{\varepsilon^{\text{MonPol}}} = \frac{b_{22}^2}{b_{21}^2 + b_{22}^2} \end{cases}}_{\text{This sums up to 1}}$$

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- *HD* allow to track, at each point in time, the role of structural shocks in driving the VAR's endogenous variables away from their steady state
- **Example** What was the contribution of oil shocks in driving the fall in GDP growth in 1973:Q4?

How to compute historical decompositions

- As an example, let's compute the *HD* of the endogenous variables for $t = 2$ in our simple bivariate VAR
- Historical decompositions can be easily understood from the Wold representation of the VAR

$$x_t = \Phi^t x_0 + \sum_{j=0}^{t-1} \Phi^j B \varepsilon_{t-j}$$

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$$x_t = \Phi^t x_0 + \sum_{j=0}^{t-1} \Phi^j B \varepsilon_{t-j}$$

- Using the Wold representation, we can write x_2 as a function of present and past structural shocks (ε^{Demand} and ε^{MonPol}) plus the initial condition (x_0)

$$x_2 = \underbrace{\Phi^2 x_0}_{init} + \underbrace{\Phi B}_{\Theta_1} \varepsilon_1 + \underbrace{B}_{\Theta_0} \varepsilon_2$$

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$$x_2 = \underbrace{\Phi^2 x_0}_{init} + \underbrace{\Phi B}_{\Theta_1} \varepsilon_1 + \underbrace{B}_{\Theta_0} \varepsilon_2$$

- Re-write x_2 in matrix form

$$\begin{bmatrix} y_2 \\ r_2 \end{bmatrix} = \begin{bmatrix} init_y \\ init_r \end{bmatrix} + \begin{bmatrix} \theta_{11}^1 & \theta_{12}^1 \\ \theta_{21}^1 & \theta_{22}^1 \end{bmatrix} \begin{bmatrix} \varepsilon_1^{Demand} \\ \varepsilon_1^{MonPol} \end{bmatrix} + \begin{bmatrix} \theta_{11}^0 & \theta_{12}^0 \\ \theta_{21}^0 & \theta_{22}^0 \end{bmatrix} \begin{bmatrix} \varepsilon_2^{Demand} \\ \varepsilon_2^{MonPol} \end{bmatrix}$$

How to compute historical decompositions (cont'd)

■ Then x_2 can be expressed as

$$\begin{cases} y_2 = init_y + \theta_{11}^1 \varepsilon_1^{Demand} + \theta_{12}^1 \varepsilon_1^{MonPol} + \theta_{11}^0 \varepsilon_2^{Demand} + \theta_{12}^0 \varepsilon_2^{MonPol} \\ r_2 = init_r + \theta_{21}^1 \varepsilon_1^{Demand} + \theta_{22}^1 \varepsilon_1^{MonPol} + \theta_{21}^0 \varepsilon_2^{Demand} + \theta_{22}^0 \varepsilon_2^{MonPol} \end{cases}$$

How to compute historical decompositions (cont'd)

- Then x_2 can be expressed as

$$\begin{cases} y_2 = init_y + \theta_{11}^1 \varepsilon_1^{Demand} + \theta_{12}^1 \varepsilon_1^{MonPol} + \theta_{11}^0 \varepsilon_2^{Demand} + \theta_{12}^0 \varepsilon_2^{MonPol} \\ r_2 = init_r + \theta_{21}^1 \varepsilon_1^{Demand} + \theta_{22}^1 \varepsilon_1^{MonPol} + \theta_{21}^0 \varepsilon_2^{Demand} + \theta_{22}^0 \varepsilon_2^{MonPol} \end{cases}$$

- The historical decomposition is given by

$$\underbrace{\begin{cases} HD_{y_2}^{\varepsilon^{Demand}} = \theta_{11}^1 \varepsilon_1^{Demand} + \theta_{11}^2 \varepsilon_2^{Demand} \\ HD_{y_2}^{\varepsilon^{MonPol}} = \theta_{12}^1 \varepsilon_1^{MonPol} + \theta_{12}^2 \varepsilon_2^{MonPol} \\ HD_{y_2}^{init} = init_y \end{cases}}_{\text{This sums up to } y_2}$$

$$\underbrace{\begin{cases} HD_{r_2}^{\varepsilon^{Demand}} = \theta_{21}^1 \varepsilon_1^{Demand} + \theta_{21}^2 \varepsilon_2^{Demand} \\ HD_{r_2}^{\varepsilon^{MonPol}} = \theta_{22}^1 \varepsilon_1^{MonPol} + \theta_{22}^2 \varepsilon_2^{MonPol} \\ HD_{r_2}^{init} = init_r \end{cases}}_{\text{This sums up to } r_2}$$

Identification Schemes

How to solve the identification problem?

■ Identification problem (recap)

- Identification $\rightarrow$ Find a B that satisfies $\Sigma_u = BB'$
- There are infinite of such B s
- In our simple example, we have to solve a system of 3 equations in 4 unknowns.
How can we do it?

How to solve the identification problem?

■ Identification problem (recap)

- Identification $\rightarrow$ Find a B that satisfies $\Sigma_u = BB'$
- There are infinite of such B s
- In our simple example, we have to solve a system of 3 equations in 4 unknowns. How can we do it? **Add a fourth equation!**
- Economic theory can help in providing the 'missing' equation
 - Make an assumption about the structure of the economy based on your beliefs (e.g. long-run monetary neutrality)
 - Try to map this assumption into an equation that involves the VAR parameters
- The additional equation is known as a **restriction**
 - That is: the additional equation restricts the set of infinite B matrices to a single one (or few ones) that are consistent with your assumption

Common identification schemes

1. Zero (recursive) contemporaneous restrictions (short-run restrictions)
2. Zero (recursive) long-run restrictions
3. Sign restrictions
4. External instruments
 - Combining sign restrictions and external instruments
 - Other (narrative sign restrictions, maximization of forecast error variance,...)

Common Identification Schemes

Zero short-run restrictions

Zero contemporaneous restrictions

- **Intuition** Identification is achieved by assuming that some shocks have zero contemporaneous effect on some of the endogenous variables
- **References** Sims (1980), Christiano, Eichenbaum, and Evans (1999)

Zero contemporaneous restrictions

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Zero contemporaneous restrictions

- **Intuition** Identification is achieved by assuming that some shocks have zero contemporaneous effect on some of the endogenous variables
- **References** Sims (1980), Christiano, Eichenbaum, and Evans (1999)
- For example, assume that monetary policy works with a lag and has no contemporaneous effects on output
- But how can we impose restrictions on the effect of a structural shock?

Zero contemporaneous restrictions

- **Solution** Impose zero restrictions on the impact matrix B

Zero contemporaneous restrictions

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- The b_{12} coefficient captures the contemporaneous effect of monetary policy on output growth

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & 0 \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

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- **Implication** We now have 3 structural parameters to estimate (instead of 4) and 3 restrictions implied by Σ_u

Zero contemporaneous restrictions

How to achieve identification?

- The system of equations implied by $\Sigma_u = BB'$ now becomes

$$\begin{bmatrix} \sigma_y^2 & \sigma_{yr}^2 \\ - & \sigma_r^2 \end{bmatrix} = \begin{bmatrix} b_{11} & 0 \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} \\ 0 & b_{22} \end{bmatrix}$$

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- This yields

$$\begin{cases} \sigma_y^2 = b_{11}^2 \\ \sigma_{yr}^2 = b_{11}b_{21} \\ \sigma_r^2 = b_{21}^2 + b_{22}^2 \end{cases}$$

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- And can be easily solved to get:

$$\begin{cases} b_{11} = \sigma_y \\ b_{21} = \sigma_{yr} / \sigma_y \\ b_{22} = \sqrt{\sigma_r^2 - \frac{(\sigma_{yr}^2)^2}{\sigma_y^2}} \end{cases}$$

Zero contemporaneous restrictions

Impact effects

- We can now derive the impact effects of shocks by simply re-writing the structural VAR as

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} \sigma_y & 0 \\ \sigma_{yr}^2/\sigma_y & \sqrt{\sigma_r^2 - \frac{(\sigma_{yr}^2)^2}{\sigma_y^2}} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

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- A one standard deviation shock to monetary policy ($\varepsilon_t^{MonPol} = 1$) in t leads to

$$\begin{cases} y_t = 0 \\ r_t = \sqrt{\sigma_r^2 - \frac{(\sigma_{yr}^2)^2}{\sigma_y^2}} \end{cases}$$

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$$\begin{cases} y_t = 0 \\ r_t = \sqrt{\sigma_r^2 - \frac{(\sigma_{yr}^2)^2}{\sigma_y^2}} \end{cases}$$

- A one standard deviation shock to aggregate demand ($\varepsilon_t^{Demand} = 1$) in t leads to

$$\begin{cases} y_t = \sigma_y \\ r_t = \sigma_{yr}^2/\sigma_y \end{cases}$$

Zero contemporaneous restrictions

Cholesky identification

- This identification scheme is normally implemented via a Cholesky decomposition of Σ_u
- A Cholesky decomposition allows us to decompose Σ_u into the product of a lower triangular matrix P times its transpose

$$\Sigma_u = PP'$$

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- This identification scheme is normally implemented via a Cholesky decomposition of Σ_u
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- In matrix form we have

$$\begin{bmatrix} \sigma_y^2 & \sigma_{yr}^2 \\ \sigma_{yr}^2 & \sigma_r^2 \end{bmatrix} = \begin{bmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} p_{11} & p_{21} \\ 0 & p_{22} \end{bmatrix}$$

Cholesky decomposition of a matrix [Back to basics]

- The Cholesky decomposition is (roughly speaking) the square root of a matrix
 - As for a square root, you can't compute a Cholesky decomposition for a non positive-definite matrix
- A symmetric and positive-definite matrix A can be decomposed as:

$$A = PP'$$

where P is a lower triangular matrix (and therefore P' is upper triangular)

- The formula for the decomposition of a 2×2 matrix is

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad P = \begin{bmatrix} \sqrt{a} & 0 \\ \frac{b}{\sqrt{a}} & \sqrt{c - \frac{b^2}{a}} \end{bmatrix}$$

Zero contemporaneous restrictions

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- To see why the zero contemporaneous restrictions identification can be implemented with a Cholesky decomposition, first note that Σ_u is a positive semi-definite matrix
- Then we can use the Cholesky decomposition to write

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$$\Sigma_u = BB'$$

- As both P and B are lower triangular, it must follow that $P = B$

Example of zero short-run restrictions

- Stock and Watson (2001). “Vector Autoregressions,” *Journal of Economic Perspectives*
- **Objective** Infer the causal influence of monetary policy on unemployment and inflation

Example of zero short-run restrictions

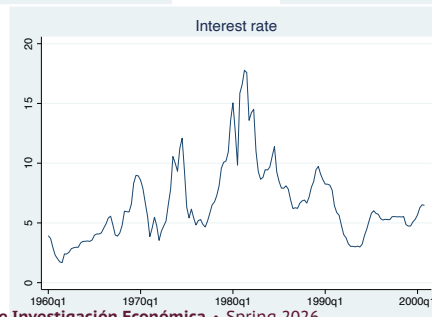
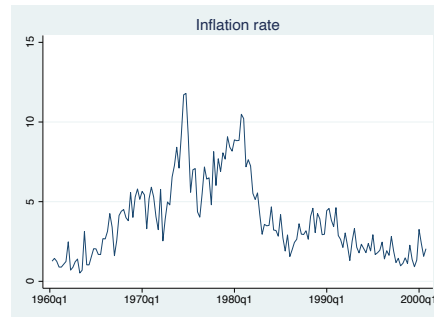
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- Assume a VAR with $p = 4$ with inflation (π_t), unemployment (u_t), and the fed funds rate (r_t)
- **Key identifying assumptions:**
 - MP (r_t) reacts contemporaneously to movements in inflation and in unemployment. The Fed considers this information when deciding the policy rate
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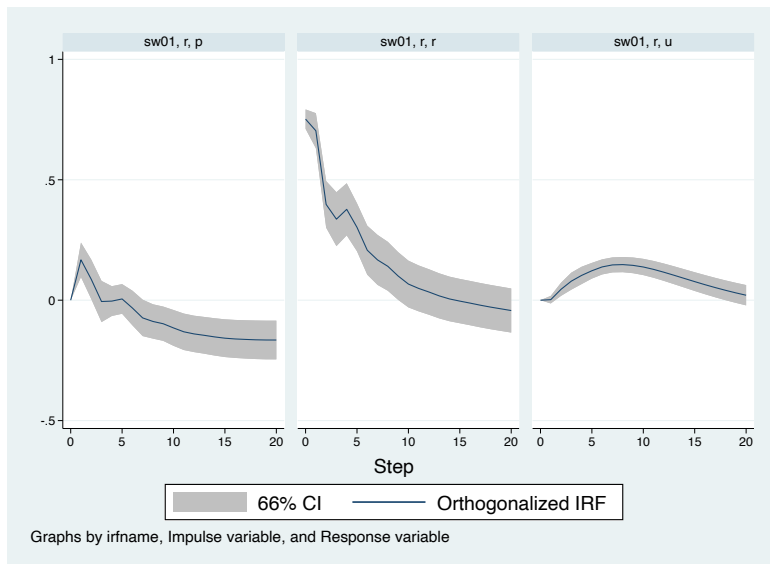
$$\begin{bmatrix} \pi_t \\ u_t \\ r_t \end{bmatrix} = \sum_{p=1}^4 \Phi_p x_{t-p} + \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} \varepsilon_t^1 \\ \varepsilon_t^2 \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

US quarterly data from 1960:Q1 to 2000:Q4



The effect of a monetary policy shock

- Monetary policy shock: $\uparrow$ inflation in the short run (price puzzle) and $\uparrow$ unemploy.



The other two shocks are identified by definition... but how can we interpret them?

- How about ε_t^1 and ε_t^2 ?
 - The shock ε_t^1 affects all variables contemporaneously
 - The shock ε_t^2 affects r_t contemporaneously but not π_t

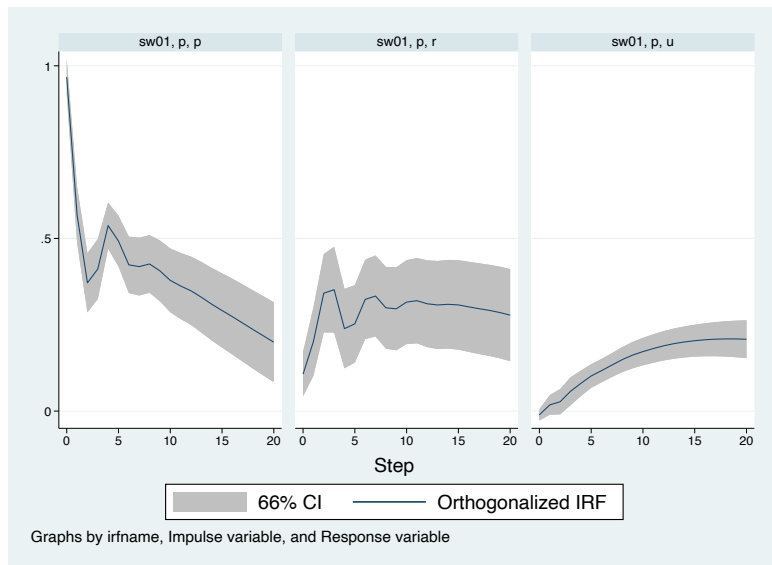
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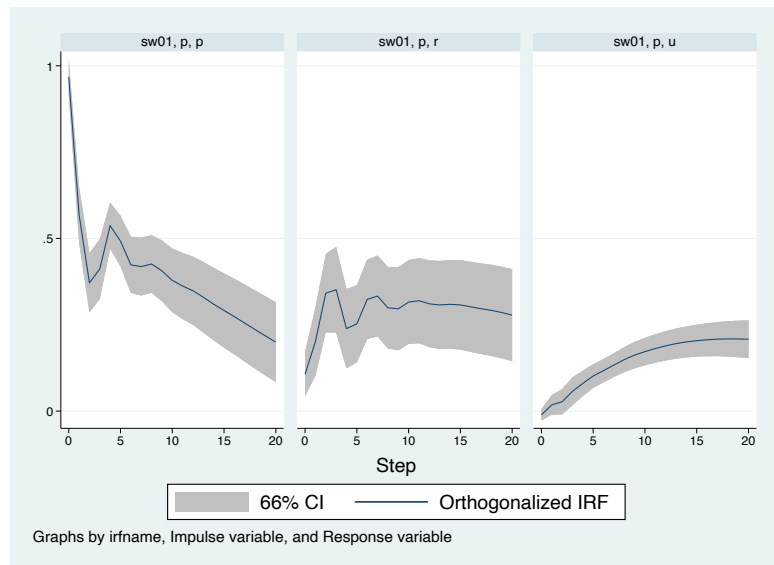
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- Some shocks may be better identified than others

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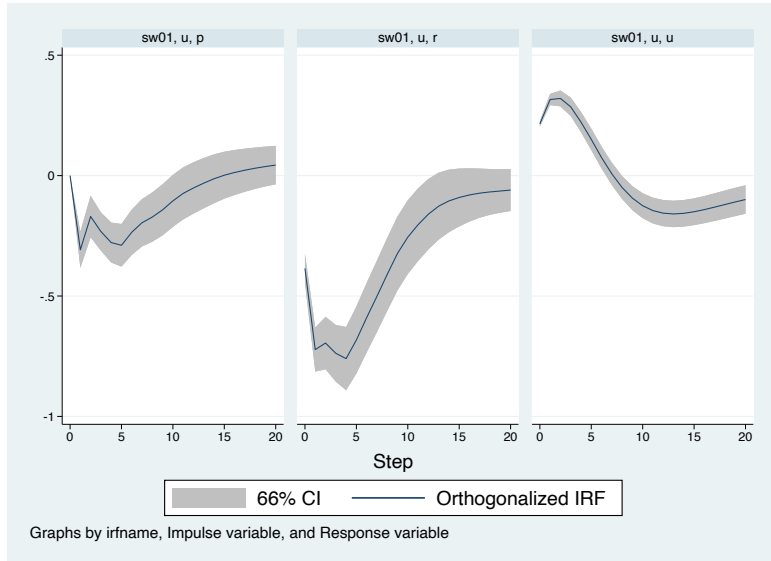


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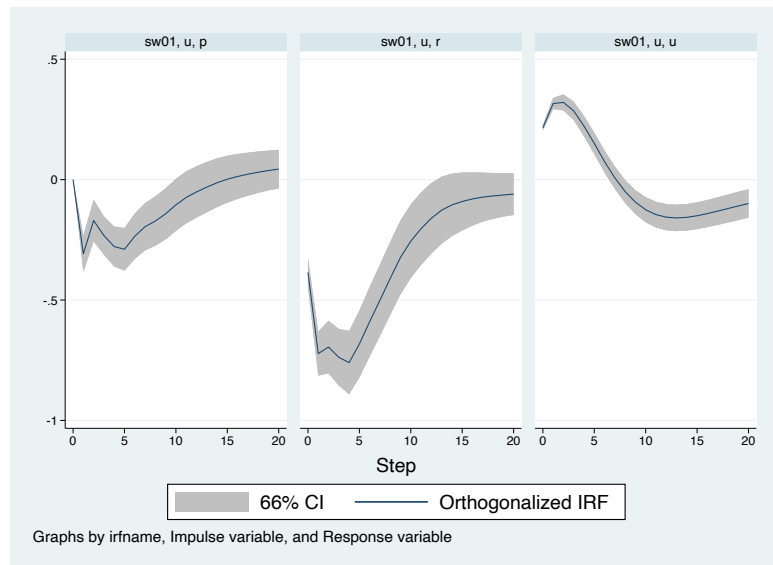


Shock to ε_t^1 behaves as a negative aggregate supply shock

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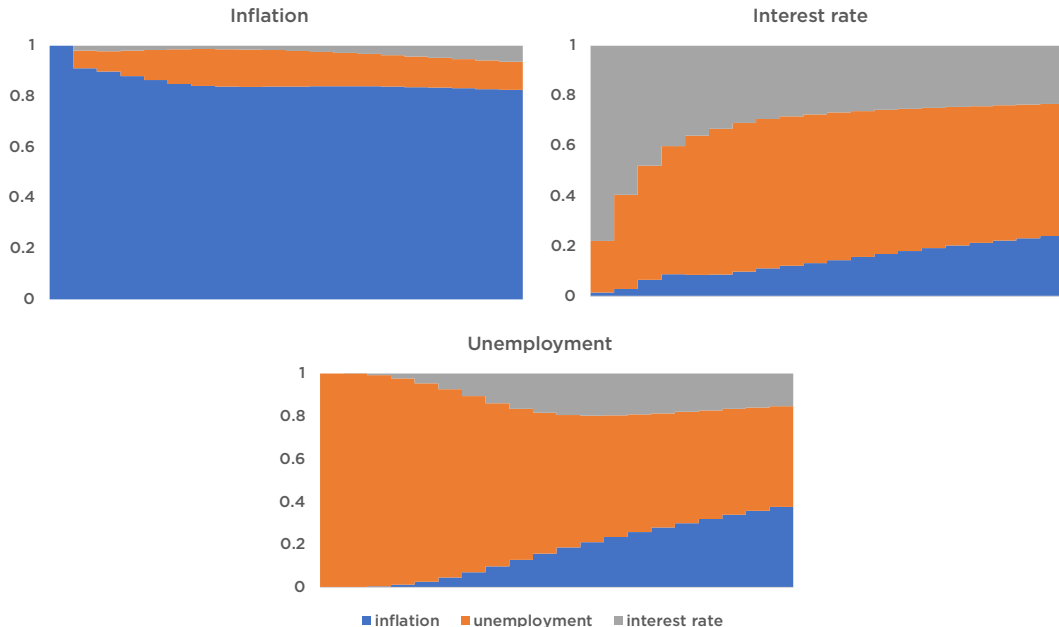


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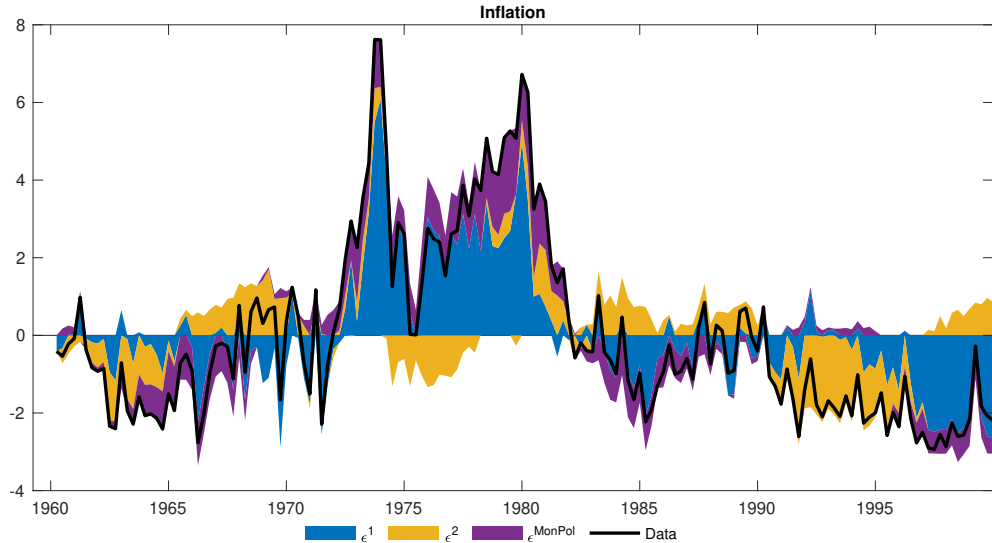


Shock to ε_t^2 behaves as a negative aggregate demand shock

Forecast error variance decomposition



Historical decomposition



Zero short-run restrictions - Limitations

- Different orderings may deliver different IRFs. **Without a structural model in mind is difficult to choose between Cholesky systems/orderings.**
- Unless some strong restrictions are imposed, dynamic models do not usually have a Cholesky structure.

Common Identification Schemes

Zero long-run restrictions

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- **References** Blanchard and Quah (1989), Gali (1999)

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- But how can we impose restrictions on the long-run cumulative effect of a structural shock?

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How to compute the cumulative long-run effects of shocks?

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- Note: for output growth, $y_{t,t+\infty}$ is the effect of ε_t on the level of output

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- If the VAR is stable, we can rewrite

$$x_{t,t+\infty} = \sum_{j=0}^{\infty} \Phi^j B\varepsilon_t = (I - \Phi)^{-1} B\varepsilon_t = C\varepsilon_t$$

where $C \equiv (I - \Phi)^{-1} B$ captures the cumulative effect of ε_t on x_t from t to ∞

Zero long-run restrictions

How to compute the cumulative long-run effects of shocks?

- What is the intuition for C ?
- Go back to our output growth / policy rate example

$$\begin{bmatrix} y_{t,t+\infty} \\ r_{t,t+\infty} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

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- Take the first equation: $y_{t,t+\infty} = c_{11}\varepsilon_t^{Demand} + c_{12}\varepsilon_t^{MonPol}$
 - The coefficient c_{12} represents the impact of a monetary policy shock (hitting in t) on the **level of GDP** in the long-run
 - If you believe in the long-run neutrality of monetary policy you would expect $c_{12} = 0$

Zero long-run restrictions

How to achieve identification?

- Remember that $C \equiv (I - \Phi)^{-1} B$ is **unobserved** as we don't know B . So, how does this help with the identification of B ?

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- To achieve identification define $\Omega \equiv CC'$ and note that
 1. Ω is known!

$$\Omega = \left((I - \Phi)^{-1} \right) B B' \left((I - \Phi)^{-1} \right)' = \left((I - \Phi)^{-1} \right) \Sigma_u \left((I - \Phi)^{-1} \right)'$$

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- We achieved identification: $B = (I - \Phi)P$

Zero long-run restrictions

How to achieve identification?

- As before, we can rewrite the structural VAR with the B matrix implied by the zero long run restriction

$$B = (I_2 - \Phi)P = (I_2 - \Phi) \times \text{chol} \left(\left((I - \Phi)^{-1} \right) \Sigma_u \left((I - \Phi)^{-1} \right)' \right)$$

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- Note that the B matrix **is not triangular**
 - This is different to what we had in the zero contemporaneous restrictions identification
- The impact effects are left unrestricted, the restrictions are on the C matrix
 - We'll check later that the restrictions is satisfied (we won't do it in class but there is a simple example with true data)

Common Identification Schemes

Sign restrictions

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 - Demand shocks should lead to an increase in output and interest rates
 - Monetary policy shocks should lead to a fall in output and an increase in interest rates

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- The matrix $B = PQ$ is a valid 'candidate' impact matrix that solves the identification problem!

- Differently from P , the matrix PQ is not lower triangular anymore

Orthonormal matrix [Back to basics]

- An orthonormal matrix Q is a real square matrix whose columns and rows are orthogonal unit vectors
- What does it mean? Take for example two 2×1 vectors q_1 and q_2 , then the matrix $Q = (q_1, q_2)$ is orthonormal if
 - The vectors have unit norm: $\|q_i\| = 1$
 - The vectors are mutually orthogonal: $q_1^T q_2 = 0$

- It follows that

$$QQ' = I \quad \text{and} \quad Q' = Q^{-1}$$

- **Note** You can draw infinite matrices that satisfy the above conditions

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- **Solution** Check that the effects of shocks implied by $B = PQ$ satisfy a set of a priori sign restrictions. That is:
 - [1] Consider the structural representation of our VAR

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

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[2] Then check that the elements of B satisfy

	Demand (ε_t^{Demand})	Monetary Policy (ε_t^{MonPol})
Output growth (y_t)	$b_{11} > 0?$	$b_{12} < 0?$
Short-rate Int. Rate (r_t)	$b_{21} > 0?$	$b_{22} > 0?$

Sign restriction in steps

■ Perform N replications of the following steps

- [1] Draw a random orthonormal matrix Q
- [2] Compute $B = PQ$ where P is the Cholesky decomposition of the reduced form residuals Σ_u
- [3] Compute the impact effects of shocks associated with B
- [4] Are the sign restrictions satisfied?
 - [4.1] Yes. Store B and go back to [1]
 - [4.2] No. Discard B and go back to [1]

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- All matrices in the set $B^{(i)}$ (for $i = 1, 2, \dots, N$) represent admissible solutions to the identification problem

- In this sense, sign restricted VARs are only **set identified**

Common Identification Schemes

External Instruments (or Proxy SVARs)

External instruments

- **Intuition** Exploit information from a variable that is *external* to the VAR, but that is correlated with a particular shock of interest and uncorrelated with other shocks (the *instrument*)
- **References** Stock and Watson (2012), Mertens and Ravn (2013)

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- For example, assume that you have some 'narrative' series of policy surprises (i.e. that are not just a response of the central bank to some development in the economy)
- But how can this help in finding the B matrix?

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- **Key element** Presence of an *instrument* that is correlated with a shock of interest and uncorrelated with all other shocks

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- For example, assume that such an instrument exists (z_t) and satisfies the following properties:

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- Then, we can identify one column (in this example, the second one) of the B matrix:

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} - & b_{12} \\ - & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

External instruments identification: How does it work?

- Recall that the reduced form residuals are a linear combination of the two structural shocks

$$\begin{cases} u_{yt} = b_{11}\epsilon_t^{Demand} + b_{12}\epsilon_t^{MonPol} \\ u_{rt} = b_{21}\epsilon_t^{Demand} + b_{22}\epsilon_t^{MonPol} \end{cases}$$

- The OLS estimate of β in the following 'first stage' regression identifies b_{22} up to a scaling factor

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- To see that, recall that the OLS β can be written as $\beta = Cov(u_{rt}, z_t) / Var(z_t)$
 - Focus on the Cov term and plug in the definition of u_{rt} to get

$$Cov(u_{rt}, z_t) = Cov(b_{21}\varepsilon_t^{Demand} + b_{22}\varepsilon_t^{MonPol}, z_t) = b_{22}Cov(\varepsilon_t^{MonPol}, z_t) = b_{22}c$$

- It follows that $\beta = \frac{b_{22}c}{Var(z_t)}$
 - As c is an unknown constant, $b_{22} = \beta Var(z_t) / c$ is only identified to a scaling factor

External instruments identification: How does it work?

- The OLS estimate of γ in the following 'second stage' regression identifies the ratio b_{12}/b_{22}

$$u_{yt} = \gamma \hat{u}_{rt} + \zeta_t = \gamma \left(\frac{b_{22}c}{\text{Var}(z_t)} \right) z_t + \zeta_t$$

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- To see that, and recalling again that $\gamma = \text{Cov}(u_{yt}, \hat{u}_{rt}) / \text{Var}(\hat{u}_{rt})$

- Focus on the *Cov* term and plug in the definition of u_{yt} and $\hat{u}_{rt}$ to get

$$\text{Cov}(u_{rt}, \hat{u}_{rt}) = \text{Cov} \left(b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}}, \frac{b_{22}c}{\text{Var}(z_t)} z_t \right) = \frac{b_{12}b_{22}c^2}{\text{Var}(z_t)}$$

- Then focus on the *Var* term to get

$$\text{Var}(\hat{u}_{rt}) = \text{Var} \left(\frac{b_{22}c}{\text{Var}(z_t)} z_t \right) = \frac{b_{22}^2 c^2}{\text{Var}(z_t)}$$

- It follows that $\gamma = \frac{b_{12}}{b_{22}}$

External instruments: Partial identification

- In sum, we can normalize the effect of ε_t^{MonPol} on r_t to 1

$$b_{22} = 1$$

- And quantify the effect of ε_t^{MonPol} on y_t as

$$b_{12} = \gamma$$

- In other words, we have identified the column of the B matrix of the structural VAR representation up to a scaling factor

$$\begin{bmatrix} y_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} - & \gamma \\ - & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_t^{Demand} \\ \varepsilon_t^{MonPol} \end{bmatrix}$$

- **Note** It is actually possible to work out the true values of B . See footnote 4 of Gertler and Karadi (2015)

Example of External instruments

- Gertler and Karadi (2015). “Monetary Policy Surprises, Credit Costs, and Economic Activity,” *American Economic Journal: Macroeconomics*
- **Objective** Infer the causal influence of monetary policy on real GDP

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- Gertler and Karadi (2015). “Monetary Policy Surprises, Credit Costs, and Economic Activity,” *American Economic Journal: Macroeconomics*
- **Objective** Infer the causal influence of monetary policy on real GDP
- Assume a VAR with $p = 12$ with industrial production, the consumer price index, the 1-year T-bill interest rate, and the Excess Bond Premium
- **Key identifying assumptions** There exists an external instrument (z_t) such that

$$\begin{aligned}\mathbb{E} \left[\varepsilon_t^i z_t' \right] &= 0 \quad \text{for } i \neq \text{MonPol} \\ \mathbb{E} [\varepsilon_t^{\text{MonPol}} z_t'] &= c\end{aligned}$$

- That is: z_t is correlated with the monetary policy shock and uncorrelated with all other structural shocks in the system

The instrument (z_t): High frequency monetary policy surprises

■ Ingredients

- Intra-daily data (τ denotes minutes)
- A monetary policy announcement on day t at time τ (e.g., FOMC decision)
- A policy indicator r (e.g., fed funds target)
- Price of futures contract on r for j days ahead $P_{t,\tau}^j = 100 - \mathbb{E}_{t,\tau}(r^j)$

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$$s_{t,\tau}^j = -(P_{t,\tau+20}^j - P_{t,\tau-10}^j) = \mathbb{E}_{t,\tau+20}(r^j) - \mathbb{E}_{t,\tau-10}(r^j)$$

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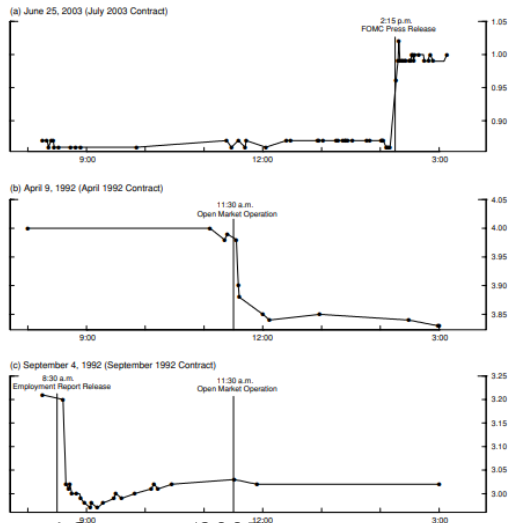
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- **Intuition** Only monetary policy shocks affect the futures prices in this short 30-minute window

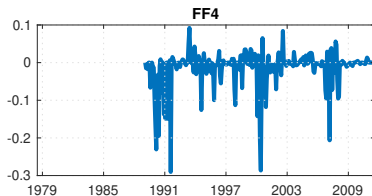
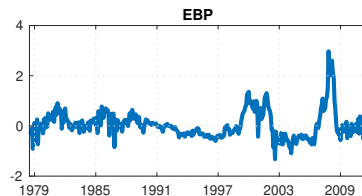
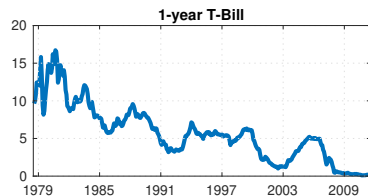
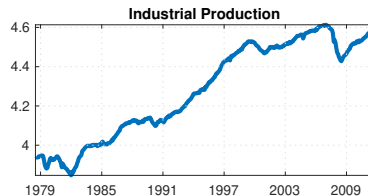
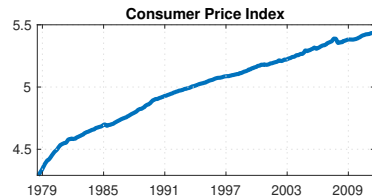
Examples of monetary policy surprises

Intra-daily change in Fed Funds futures prices around monetary events



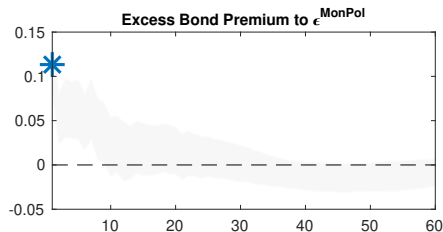
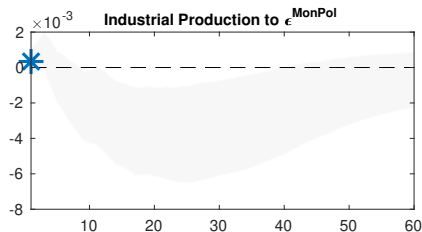
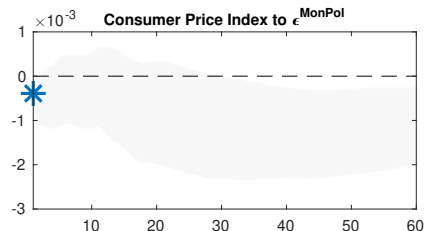
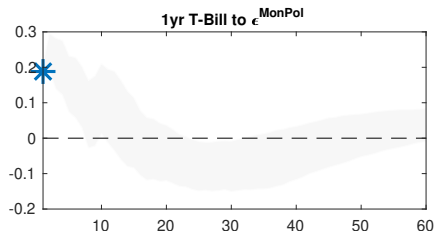
Source: Gurkaynak, Sack and Swanson (2005).

Gertler and Karadi, 2015: US monthly data, 1979:M7-2012:M6



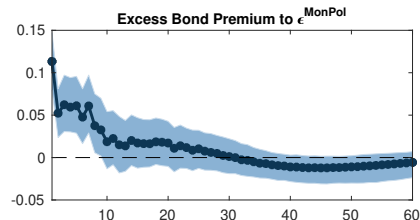
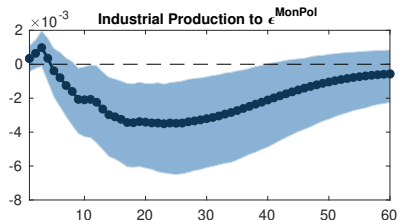
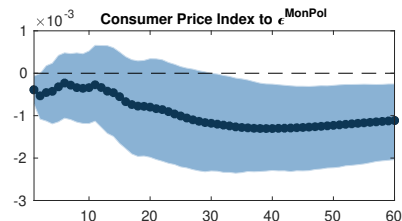
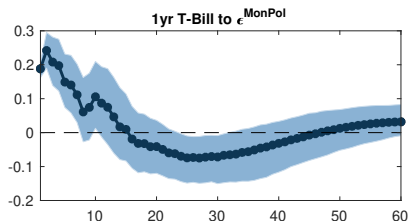
Impulse response functions: Impact effect

- The impact effect (i.e. the b matrix) is given by the first and second stage regressions

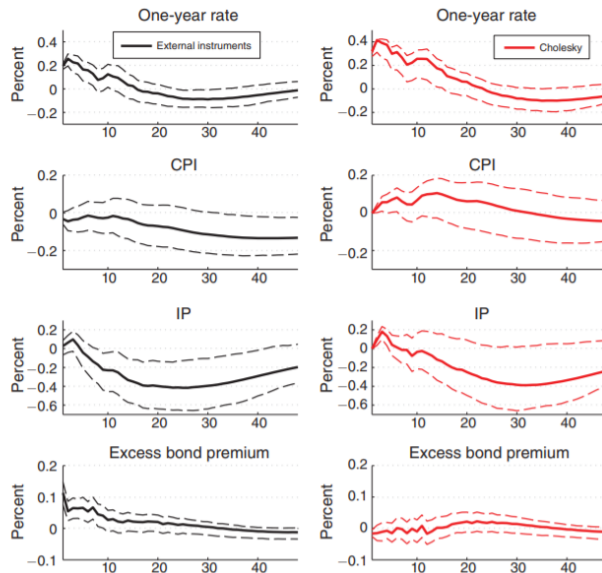


Impulse response functions: Dynamic effect

- The dynamic effect is computed as usual with the Φ matrix



External instruments vs. Cholesky decomposition



First-stage regression:

F: 21.55; Robust F: 17.64; R^2 : 7.76 percent; Adjusted R^2 : 7.40 percent

External Instruments

- Identification with external instruments does not require to impose any restriction on the response of variables
- It is particularly useful to identify the effects of the shocks on financial variables
- Main limitation: it can be hard to find proxies for the shock of interest
- It has been mostly limited to event-studies